# The Future In .NET MAUI

David Ortinau | @davidortinau



MODERN CLIENT

# David Ortinau

## Principal Product Manager

St. Louis, Missouri USA

Help developers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform (.NET MAUI).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.

# Build anything with a unified platform



# .NET Momentum

**6.1+ Million**

MONTHLY ACTIVE USERS

**53,000+**

COMMUNITY MEMBERS HAVE CONTRIBUTED TO .NET

## Top 3

**Most Loved/Admired Developer Framework (2019-2023)**

.NET Core/.NET

stackoverflow

## Top 5

**Highest velocity OSS project**

github.com/dotnet

CLOUD NATIVE COMPUTING FOUNDATION

## Top 5

**Language on GitHub**

C#







Irth

Actionable insights from the industry's most experienced and trusted.


9:53
95%
Irth


Retail Insight


hogeschool
vives

irth
UtiliSphere

S


Xari

OE
XA

MAHIZ
grow your business

H3Y

ups

CARDIFF
CAERDYDD

Fidelity

Microsoft
Dynamics 365

Microsoft
Azure

CIVICA

SPORTS
NEXT
SPORTS×TECH

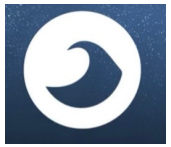
DevInMotion

Belgian
Red Cross
Flanders

ambientindustries

A

insight

esri


Soluzione 1

REY
TECHNOLOGY

[A]


# .NET MAUI at Microsoft

play.google.com

Google Play Games Apps Movies & TV Books Kids

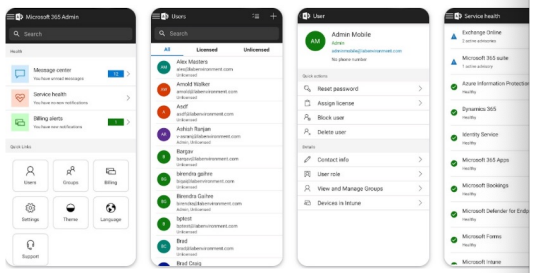
## Microsoft 365 Admin

Microsoft Corporation

4.6★ 24.5K reviews | 1M+ Downloads | Everyone

Install

This app is available for your device



About this app →

The Microsoft 365 Admin App enables you to be productive from
add users, reset passwords, manage devices, create support requ

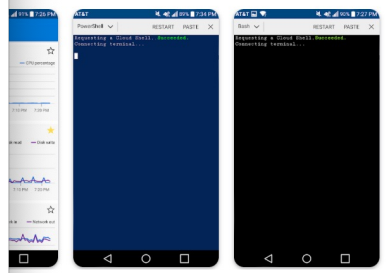
play.google.com

Google Play Games Apps Movies & TV Books Kids

## Microsoft Azure

Microsoft Corporation

4.6★ 7.16K reviews | 1M+ Downloads | Everyone



while on-the-go:
s anytime, anywhere

play.google.com

Google Play Games Apps Movies & TV Books Kids

## Store Commerce

Microsoft Corporation

4.2★ 29 reviews | 5K+ Downloads | Everyone

Install

You don't have any devices

# .NET MAUI Momentum

## Top 2

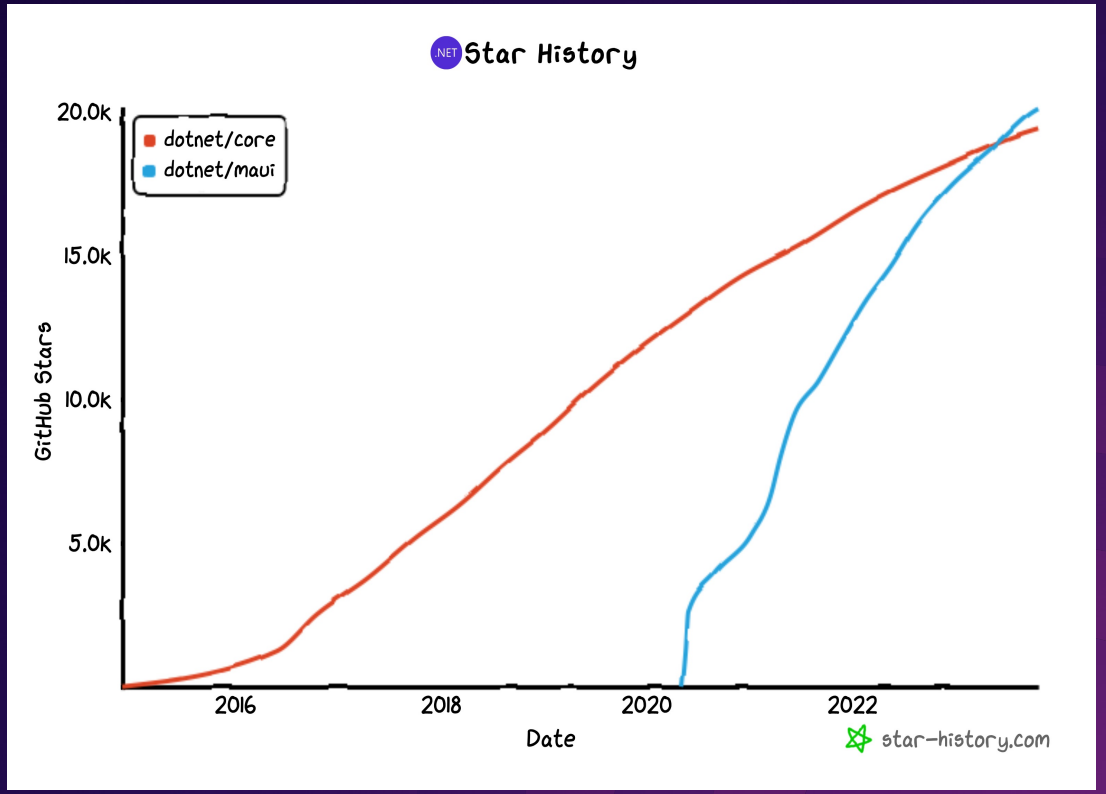
**Active GitHub users in dotnet**

## Top 2

**Most active in .NET Foundation**

## Top 5

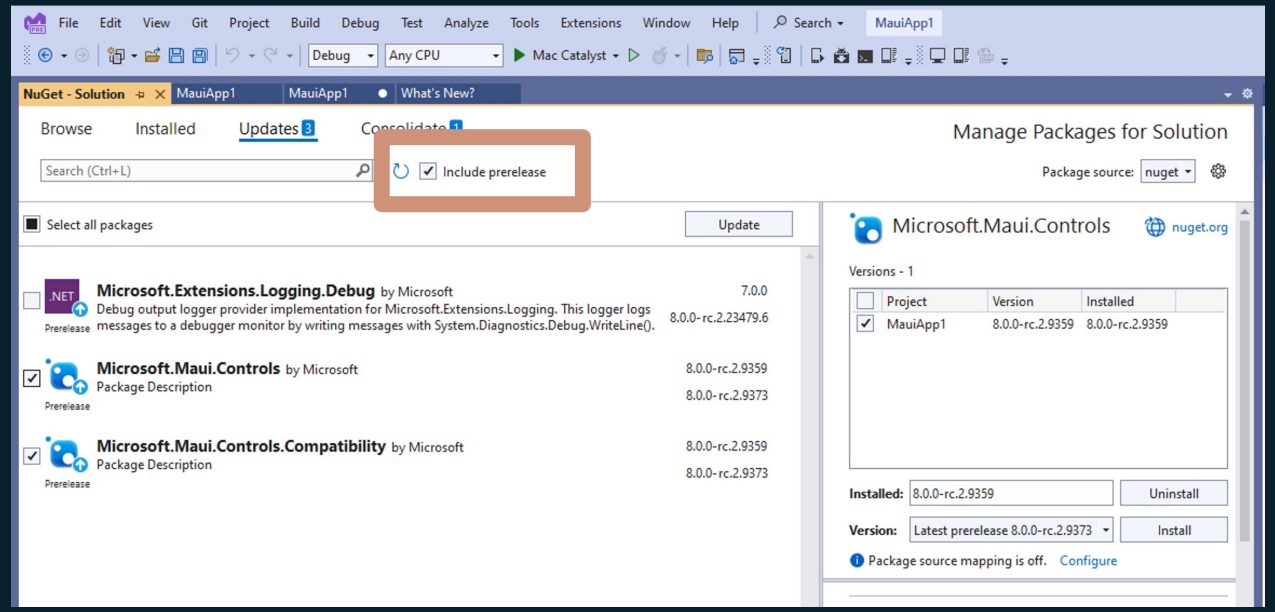
**Most active C# repo on GitHub past month**



# .NET MAUI in .NET 8

Quality, Stability, Desktop completeness

# How to Preview .NET MAUI Releases




$(MauiVersion) is read from your installed .NET workload.

You can override this in your csproj by adding <MauiVersion></MauiVersion>.

Installing from NuGet will update your PackageReference, but not <MauiVersion> if you manually added that to your csproj.

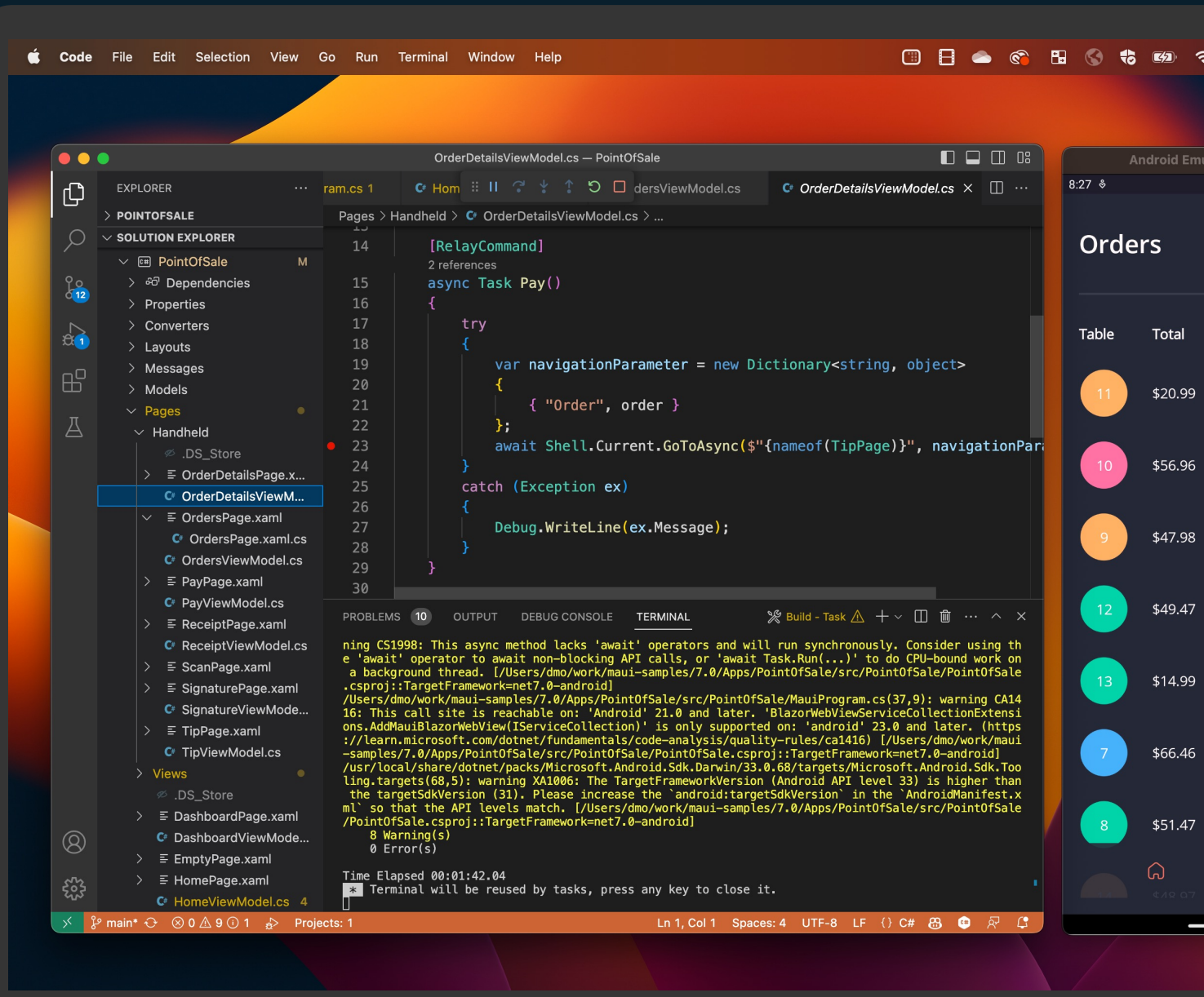
```
<ItemGroup>
  <PackageReference Include="Microsoft.Maui.Controls"
                    Version="8.0.0-rc.2.9373" />
  <PackageReference Include="Microsoft.Maui.Controls.Compatibility"
                    Version="8.0.0-rc.2.9373" />
</ItemGroup>
```

(Preview)

# .NET MAUI for Visual Studio Code

## Highlights:

- Solution explorer
- New project creation
- Environment configuration
- Debug Android, iOS, macOS, Windows
- Intellisense and IntelliCode

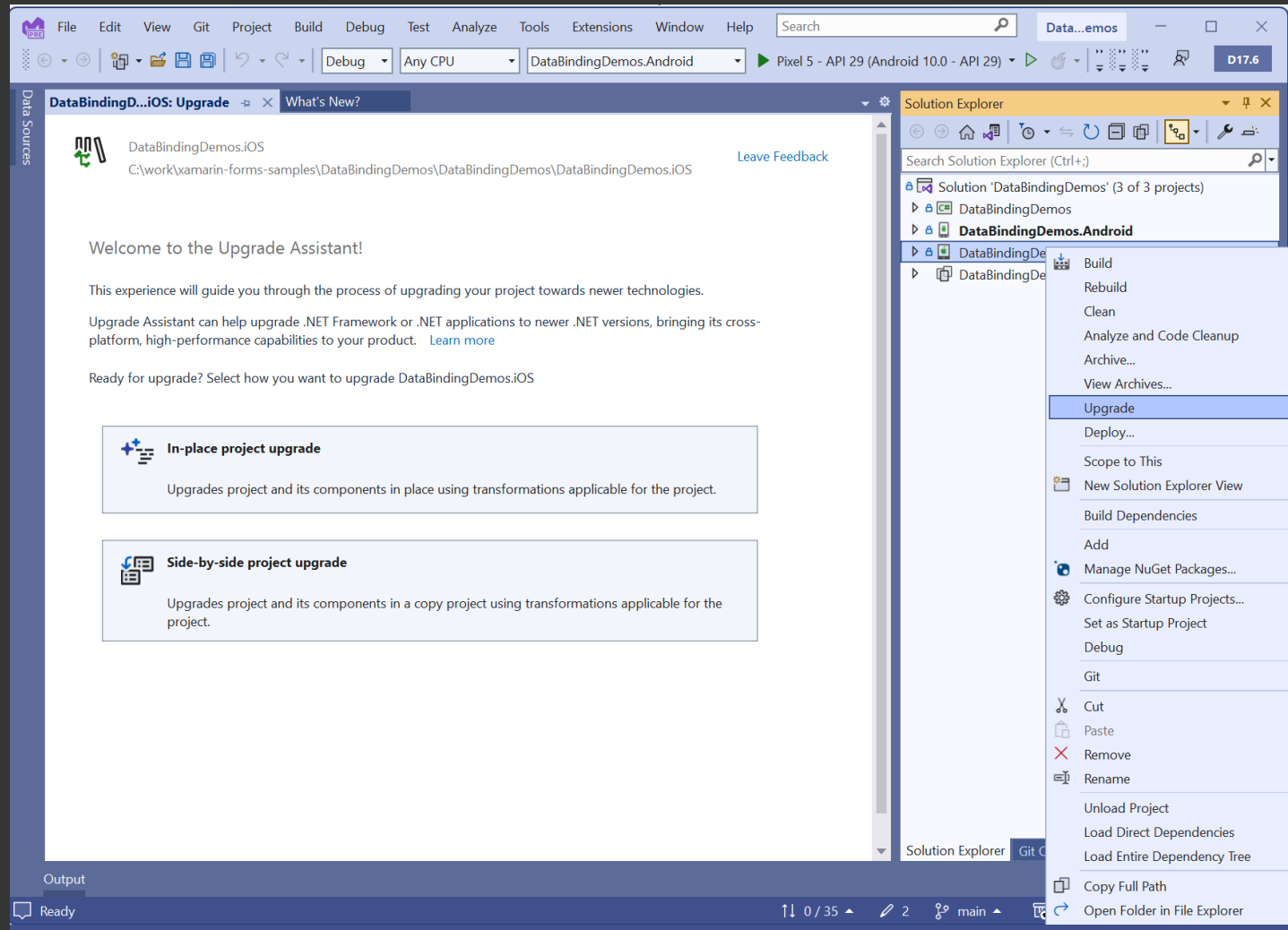


# .NET Upgrade Assistant for Xamarin Projects

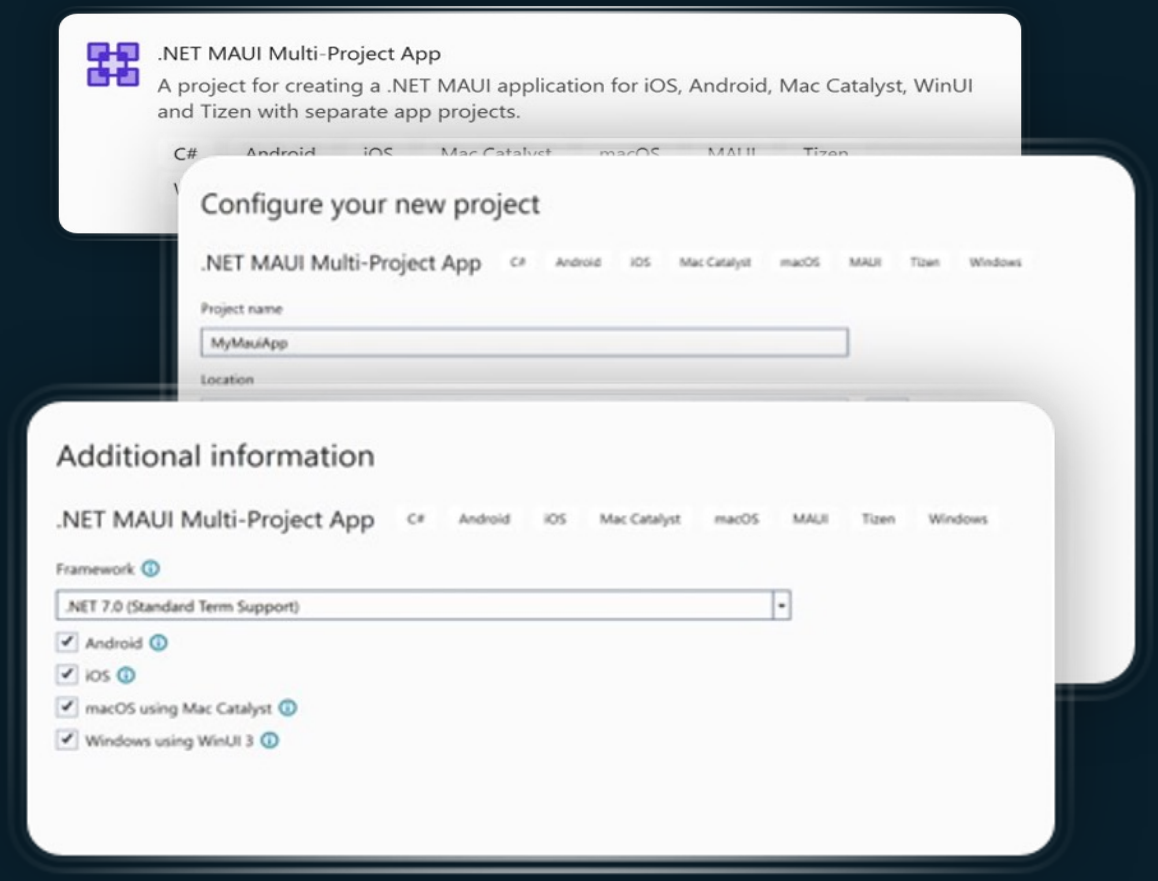
## Highlights:

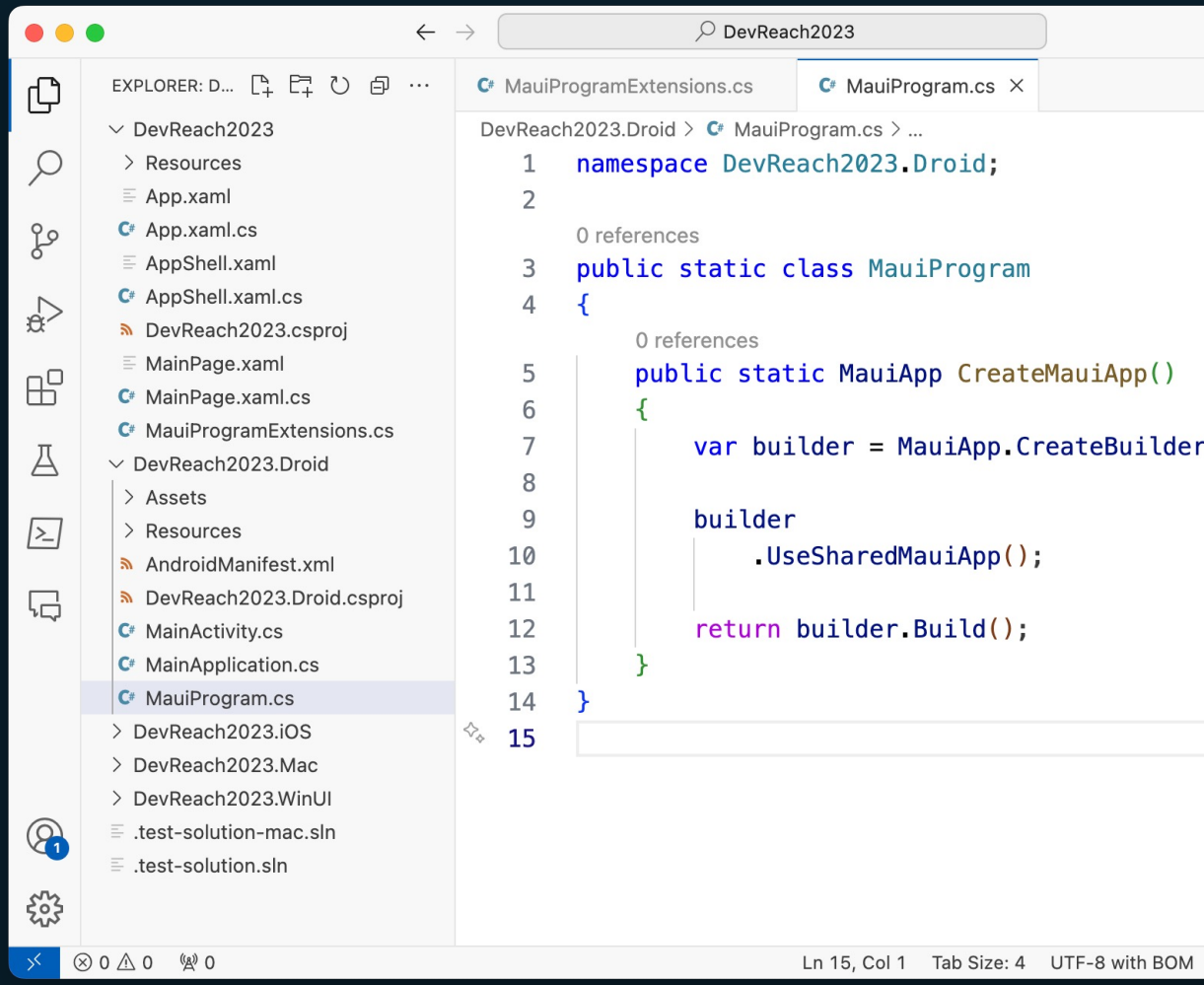
- Converts projects
- Replaces known NuGets
- Fixes namespaces
- Fixes base types
- Applies various code updates

maui-upgrades@microsoft.com



# Multi-headed MAUI apps

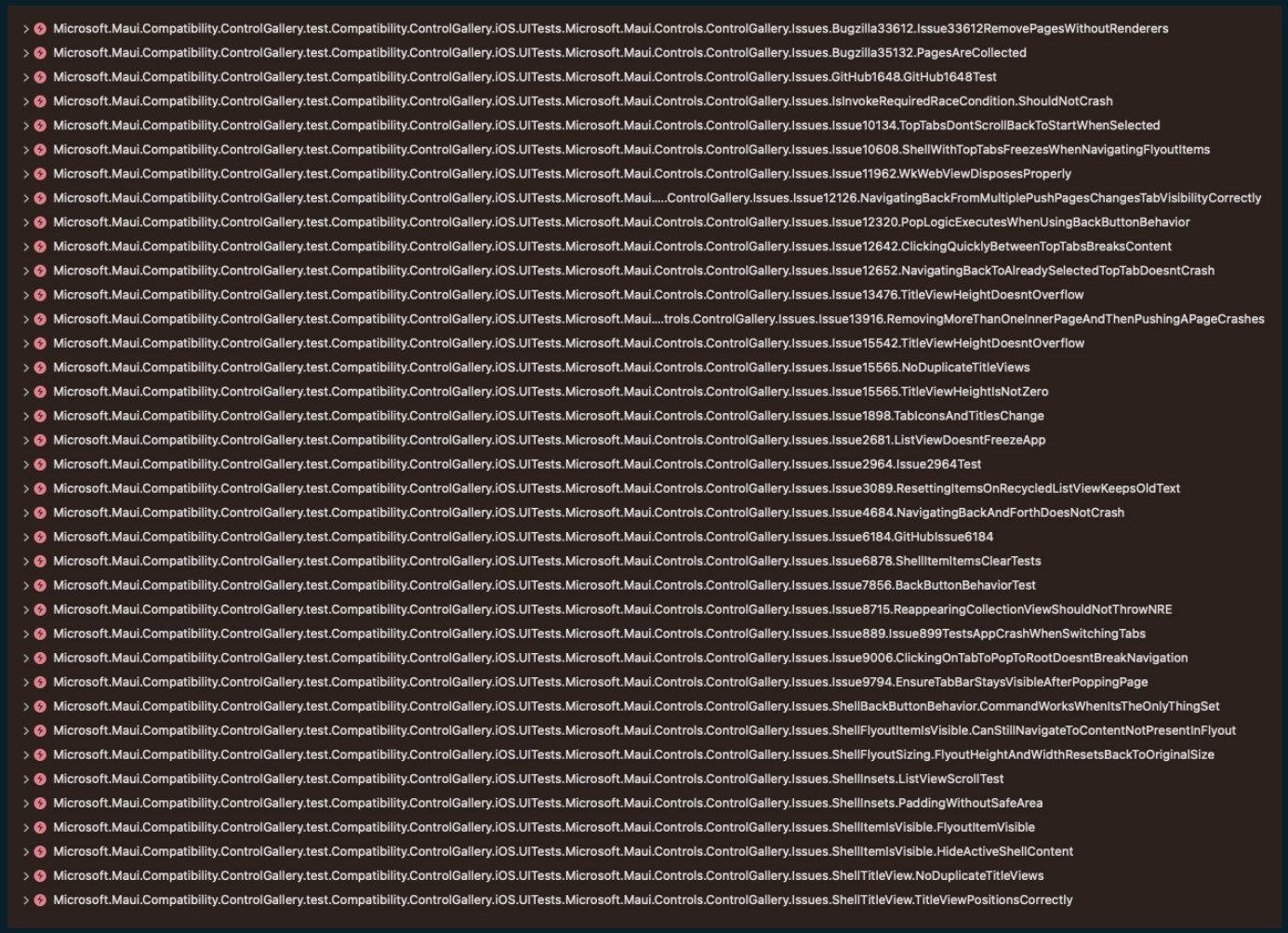


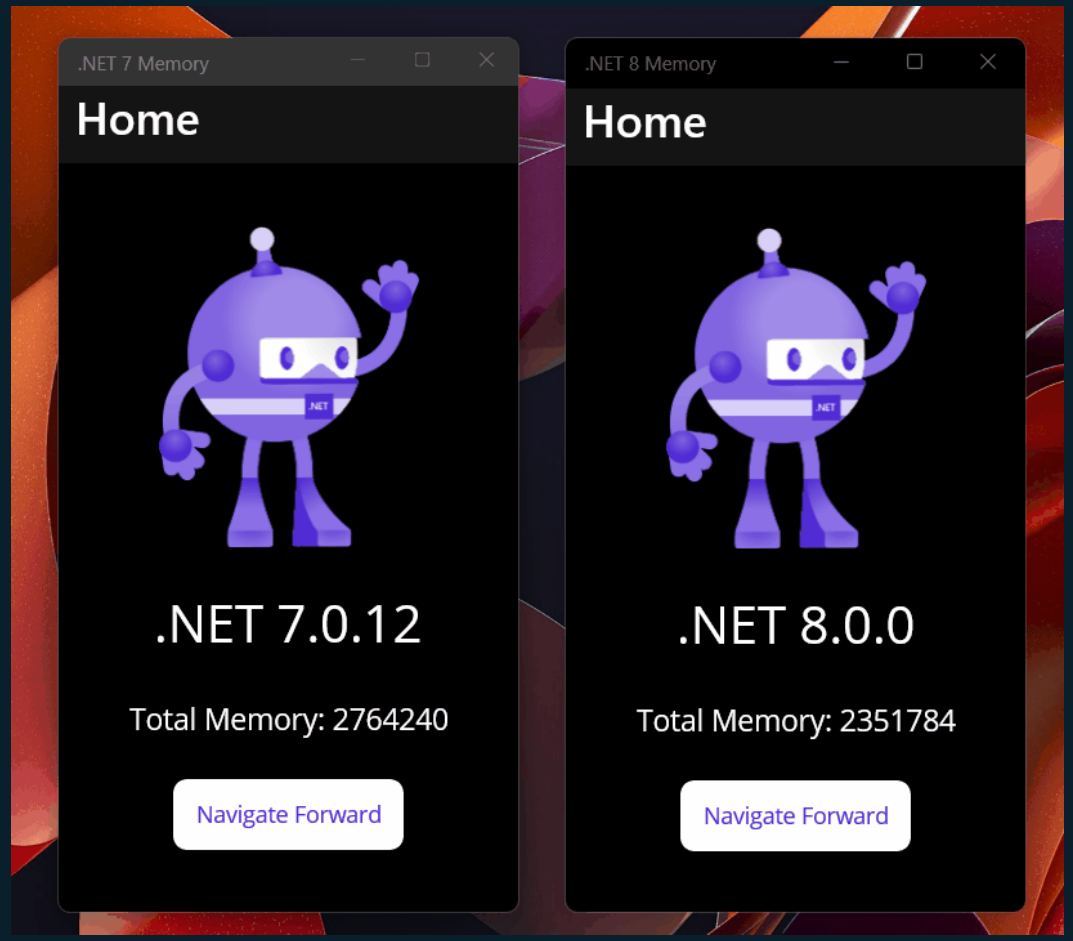

https://github.com/mattleibow/MauiMultiHeadProject/

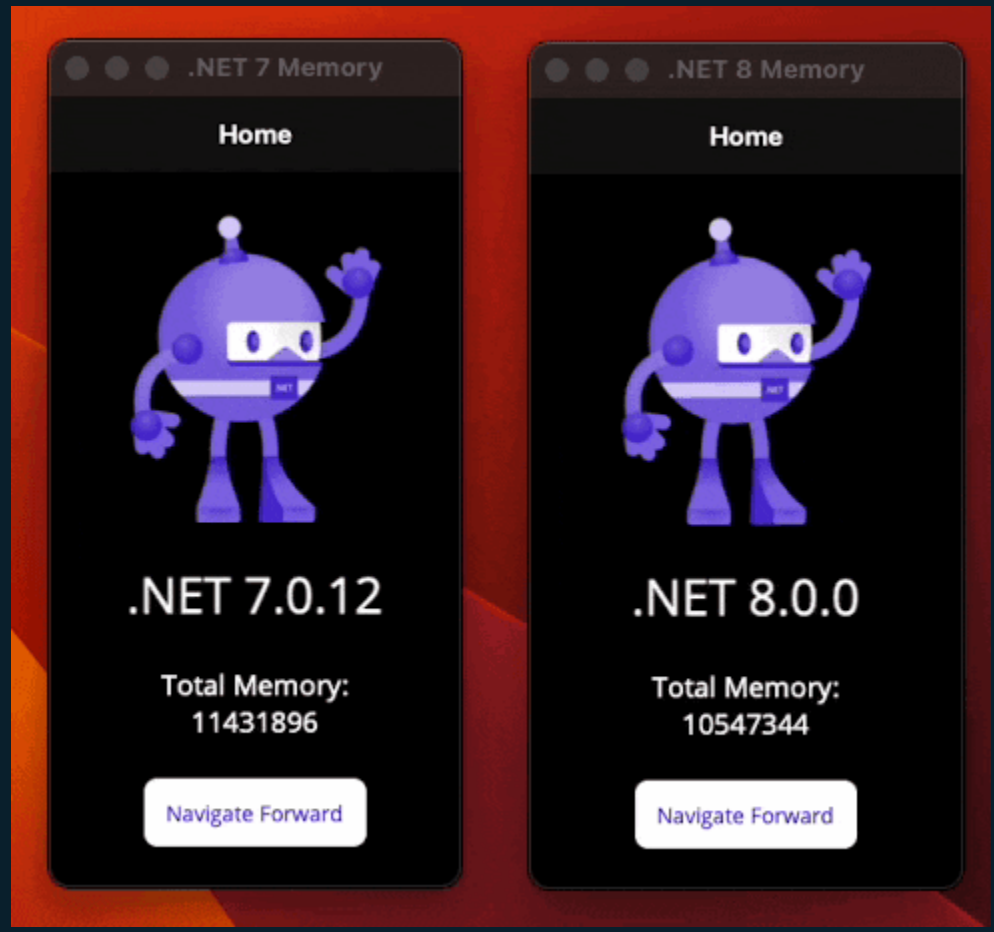
# Migration Compatibility

- Backported Legacy Tests from .NET8 to .NET7 to validate NET7 SR8
- Verified these failing tests weren't regressions between .NET7 releases.
- These are all tests that are now passing in .NET8
- This represents tests that were written against Xamarin.Forms that are now passing in .NET8 but weren't in .NET7



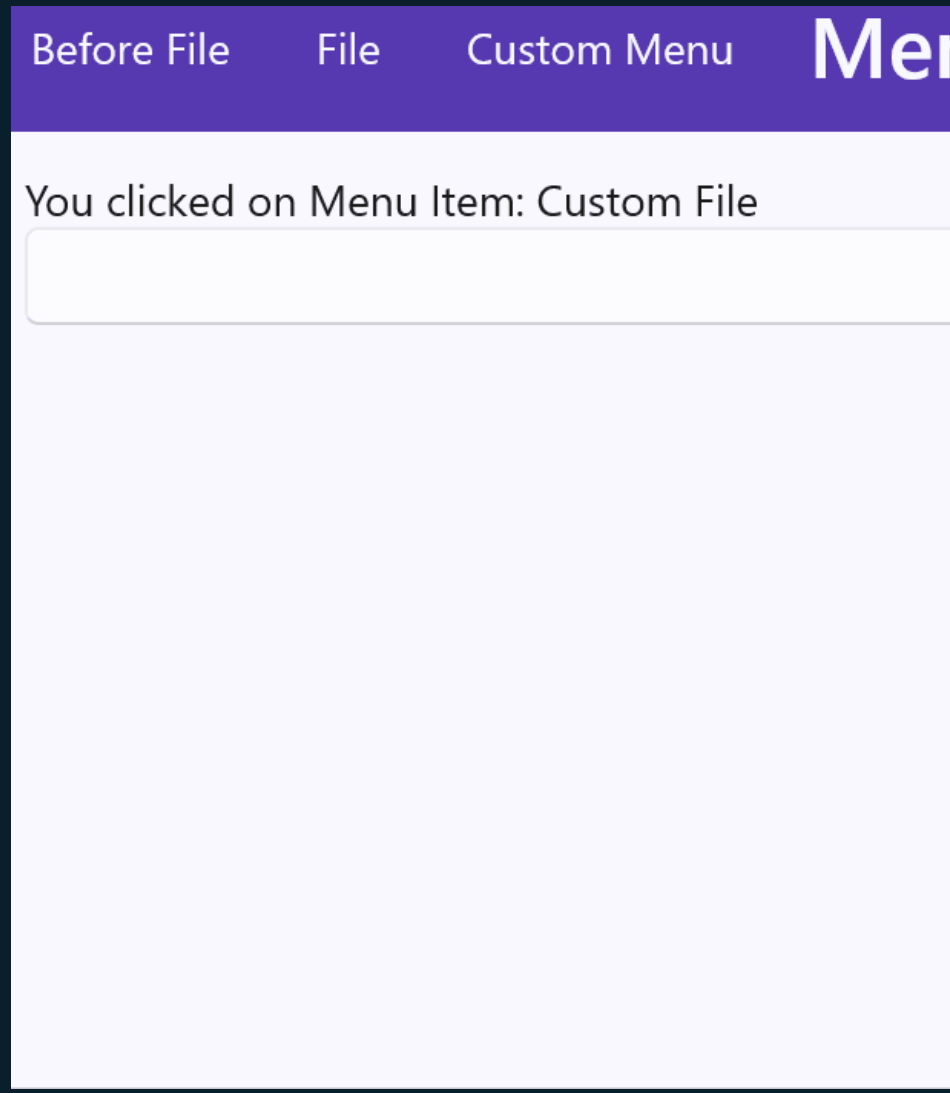
# GC.GetTotalMemory(): .NET 7 vs .NET 8



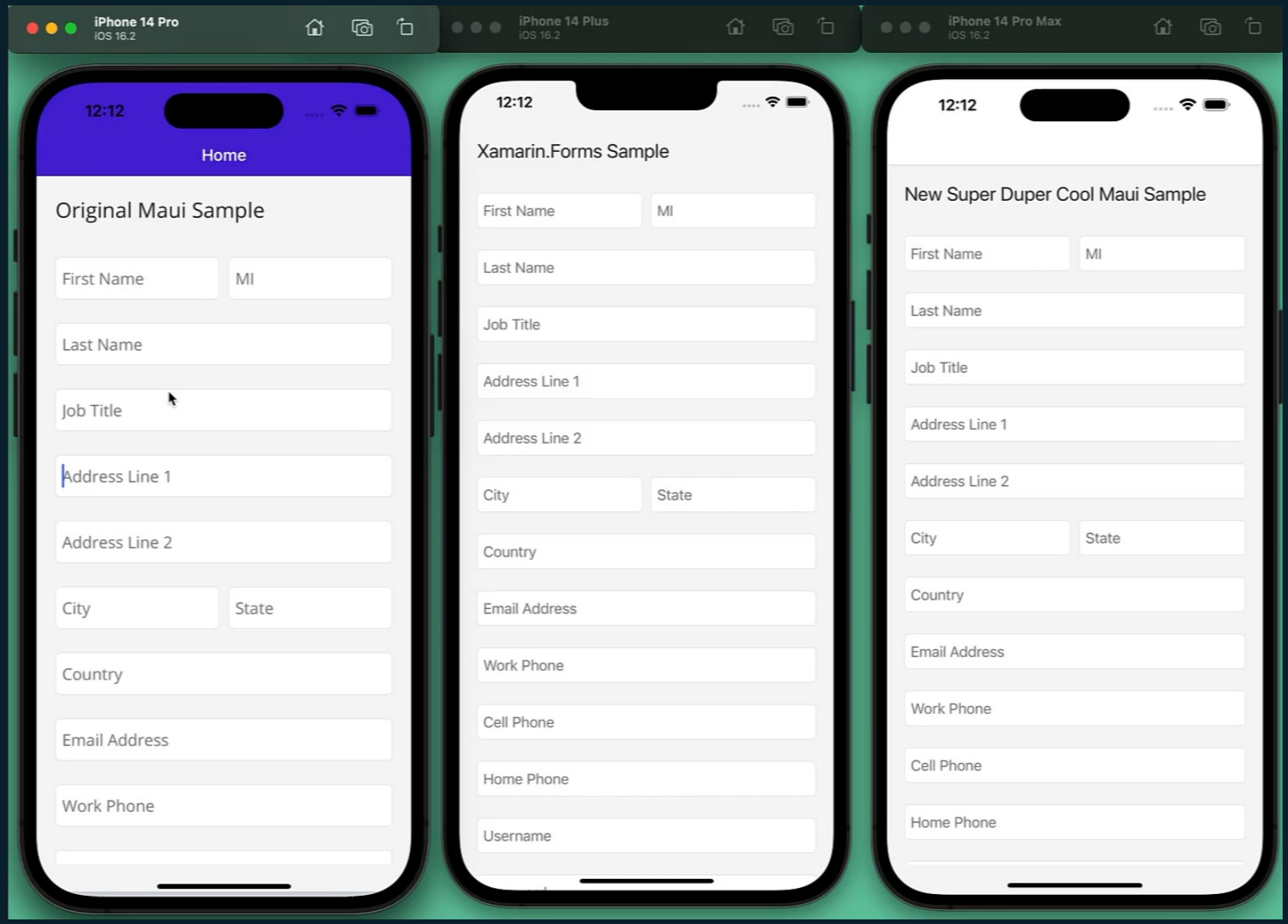
# Keyboard Accelerators

```
<MenuBarItem Text="File">
  <MenuFlyoutItem
        Text=”Custom File"
        KeyboardAccelerators=”Ctrl+F"/>
</MenuBarItem>
```



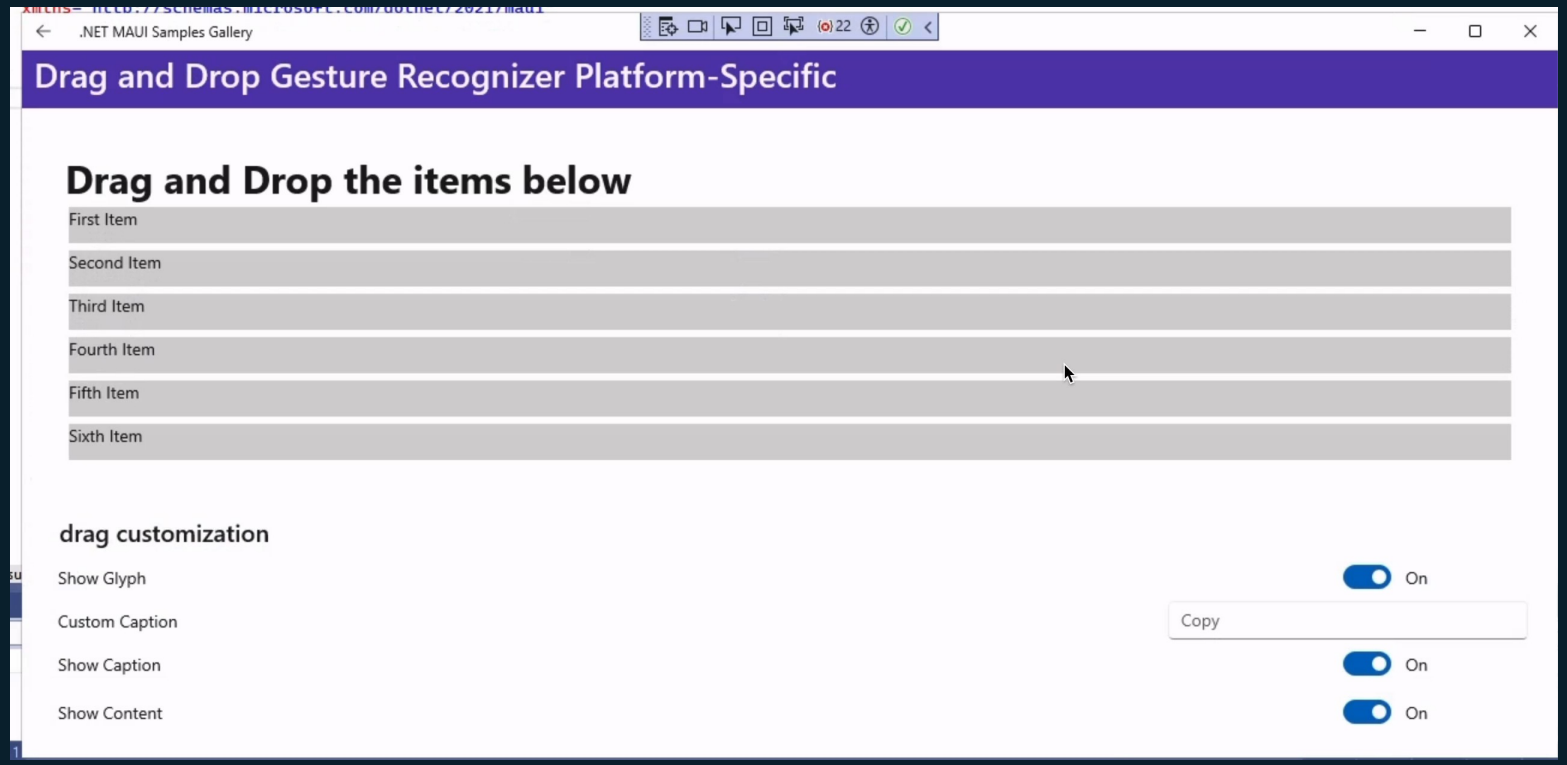
# Keyboard Improvements

- No keyboard overlap
- Scroll to focus
- Tabbing/Next defaults



# Drag & Drop

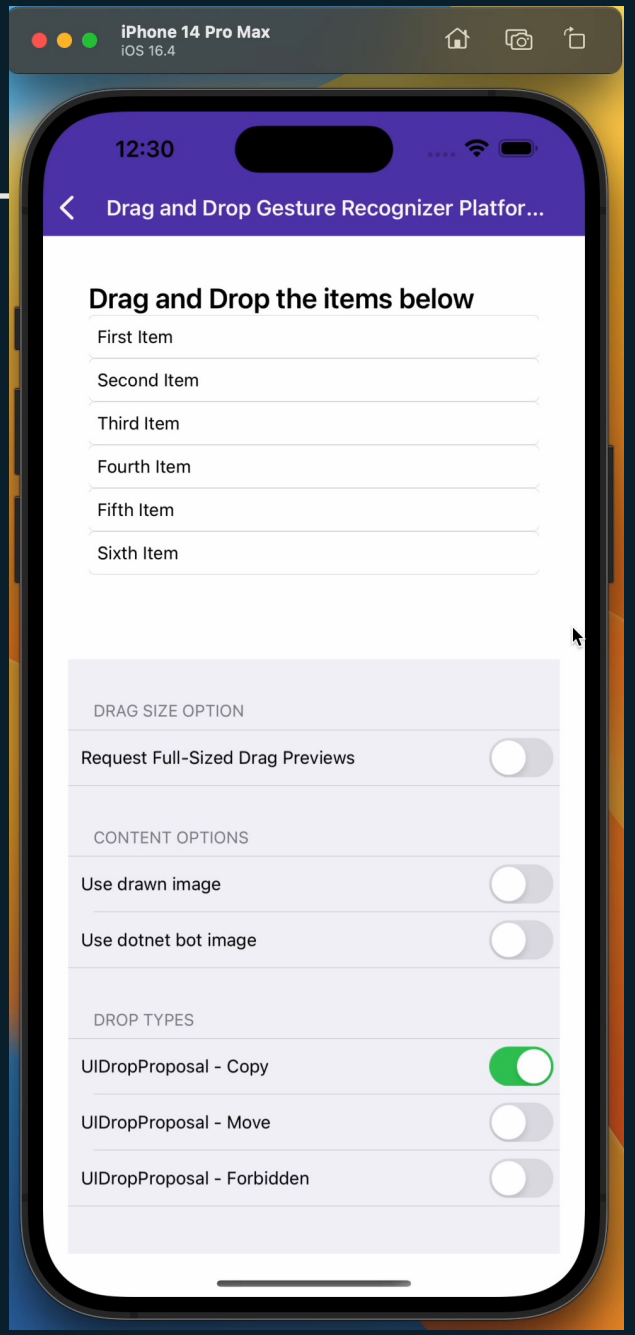
Exposed WinUI Drag & Drop APIs such as custom glyphs (images) on drag, custom captions on drag



# Drag & Drop

Exposed iOS Drag & Drop APIs such as:

- Drag items retaining their original size.
- Change the shape or image of a drag item.
- Changing the UIDropProposals (Copy, Move, Forbidden)



# Pointer Gestures

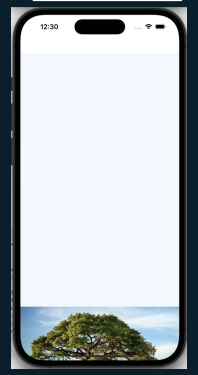
- Introduced PointerPressed and PointerReleased gesture states that detect mouse up and mouse down

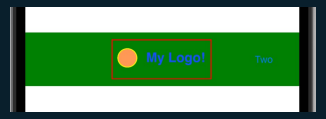


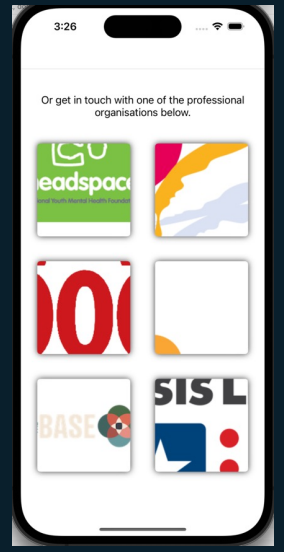
# iOS Large Images Fixed

- Downscaled images on iOS didn't respect the user's dimension constraints
- Problem caused by native iOS limitation
- Large images also overflowed their container with certain Aspect settings
- PR fixed 8 identified issues

**Before**

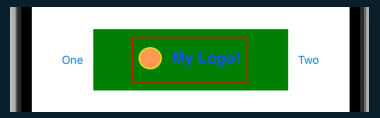


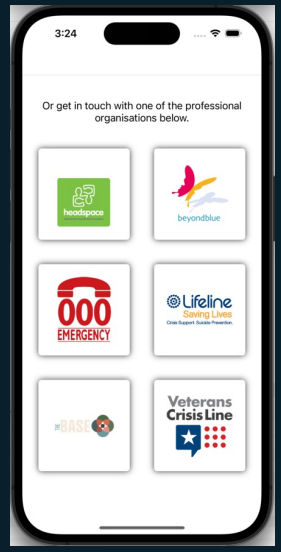




**After**

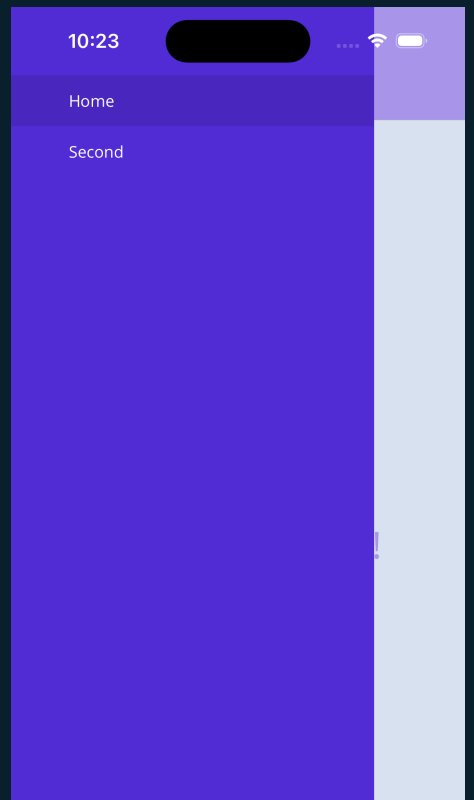


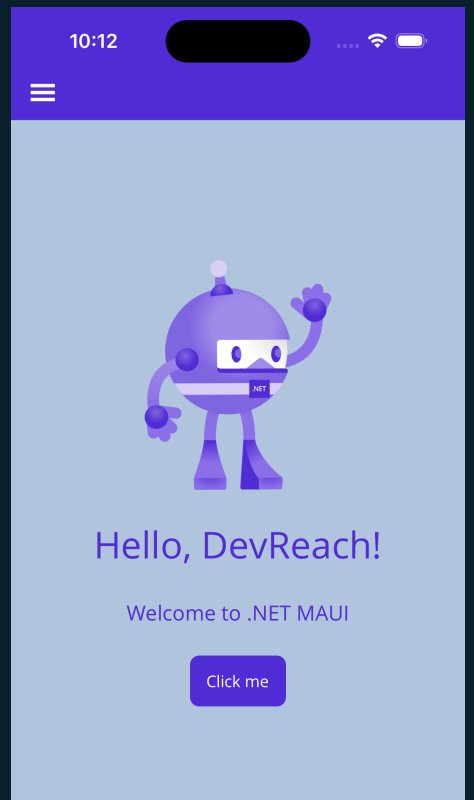



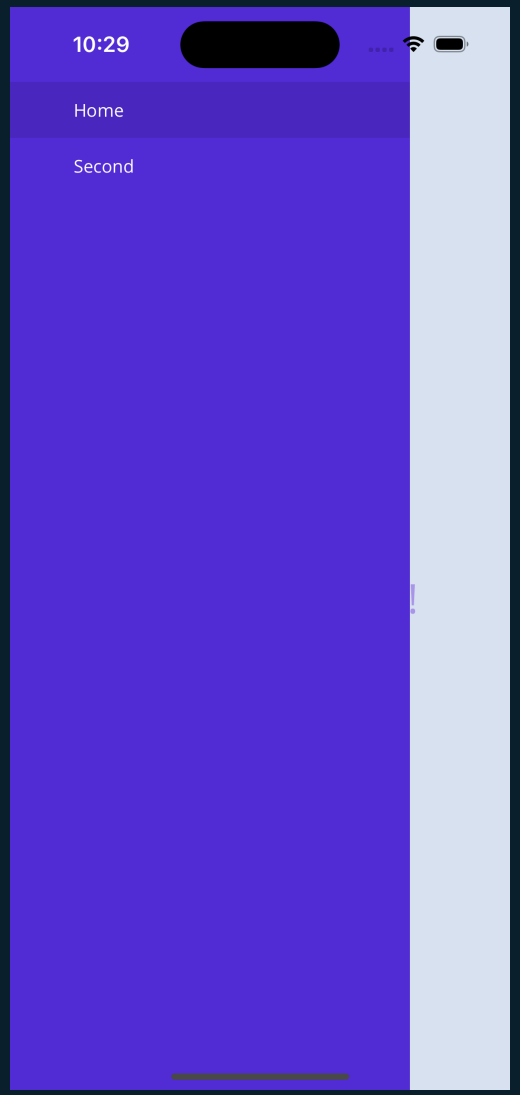



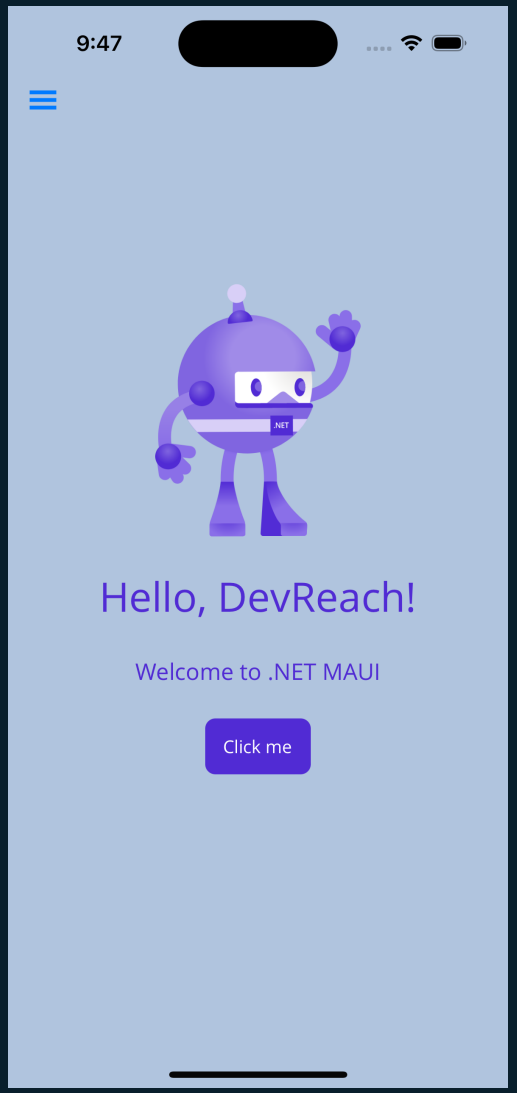
# iOS Safe Area Fixes (Shell)

ios:Page.UseSafeArea=false
IgnoreSafeArea=true

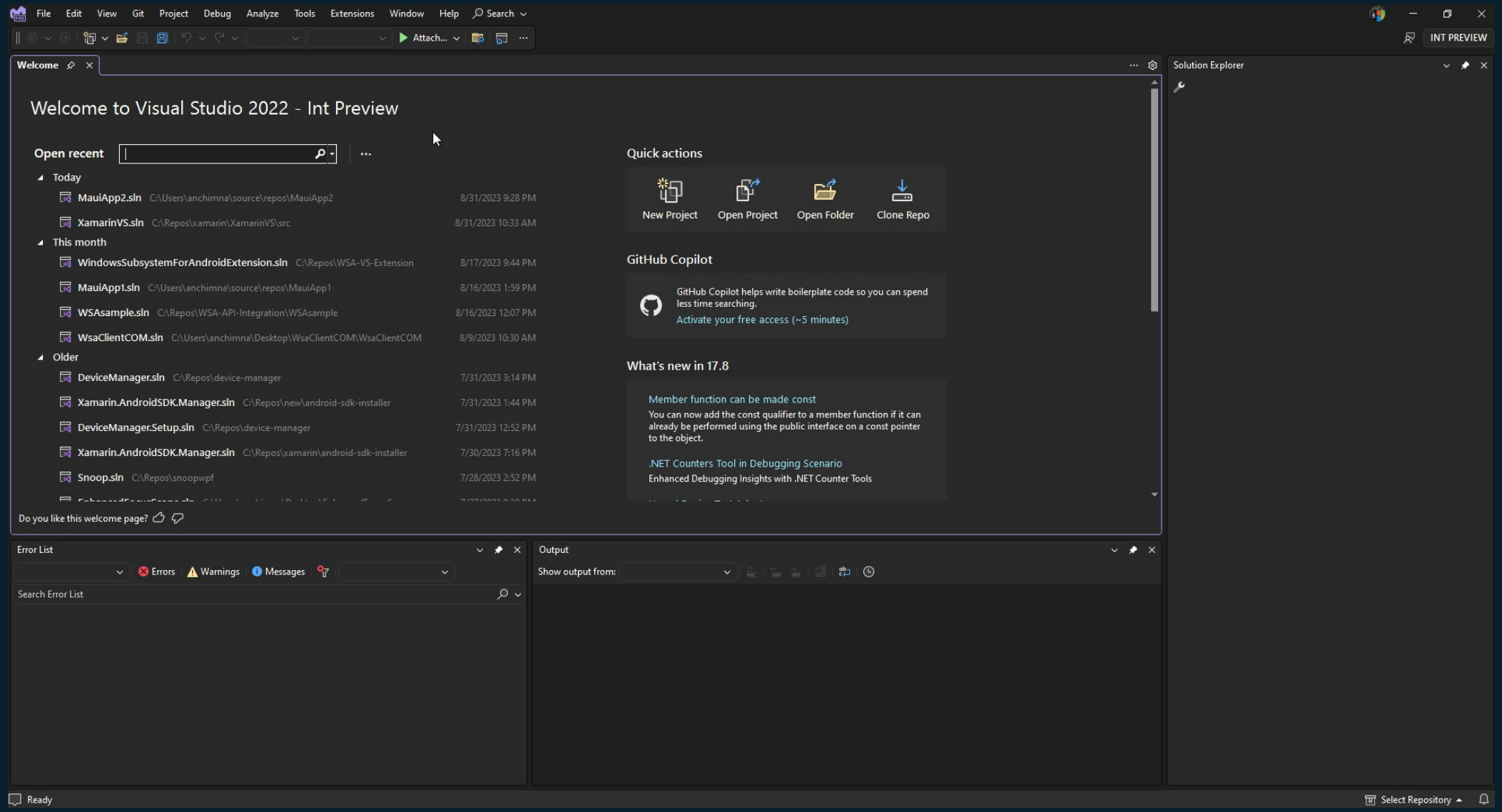


# WSA is now a debug target



# Miscellaneous .NET 8 Work

iOS 17 & Xcode 15

Android 34

WinAppSDK 1.4.2

## Removed Rosetta dependency

- Smaller install footprint
- Faster all things Apple on ARM machines

## Android.Resources.Designer

- 8% assembly size reduction
- 2% package size reduction
- 8% faster app startup

# .NET MAUI in .NET 9

Early plans

# Swift Interop

https://github.com/dotnet/designs/pull/302

213 iOS frameworks

- 122 Obj-C
- 58 Obj-C + Swift
- 33 Swift only

Swift only examples:

- CarKey
- Charts
- CryptoKit
- MusicKit
- RealityKit
- WeatherKit
- WidgetKit
- WorkoutKit
- …and more

# Misc Goals

Increase velocity of bug fixing

Reduce build times

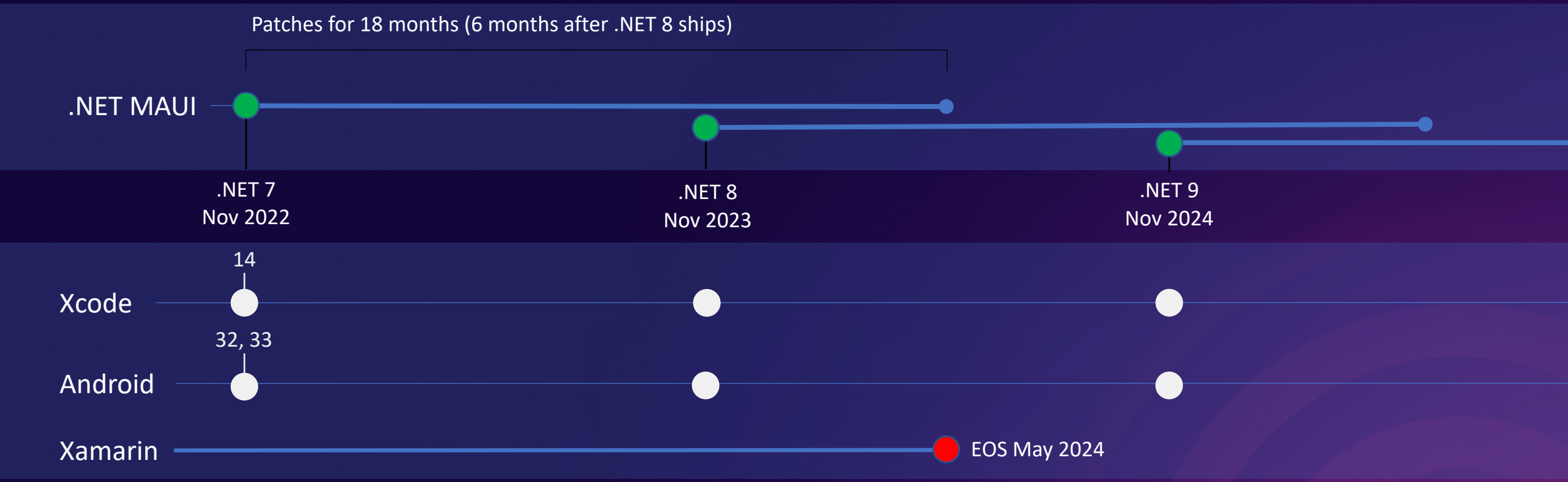
Improve runtime performance

Support more native desktop experiences

Improve hot reload

Add documentation and samples

# Release schedule, support & patching




- A major version of .NET MAUI will ship with .NET going forward (.NET 7, .NET 8, etc.)
- .NET MAUI is an optional workload with external dependencies (such as Xcode & Android SDK tools)
- A major version of .NET MAUI will receive patches for 6 months after a successor (major version) ships
- Xamarin SDKs will receive support until May 2024

# Thank You!

Let's build amazing apps with .NET 7
get.dot.net/7

